OpenEdge Bridge — Documentation

Sample INSERT Data

MySQL — `id_print` (10 records)

```
INSERT INTO id_print (batch_no, batch_date, app_no, dob, cp_no,
fullname, hcp_fullname, address, hcp_cpno, lname, fname, mname,
spa_docno, page_no, book_no, series_no, spa_atty, spa_dt_issue, printed,
printed_dt, printed_tm, id_type, id_code, print_count)
VALUES
(1, '2026-01-15', 1001, '1990-05-20', 'CP-001', 'Juan Dela Cruz', 'Dr.
Maria Santos', '123 Rizal St, Manila', 'HCP-001', 'Dela Cruz', 'Juan',
'Santos', 'SPA-001', 'P001', 'BK001', 'SR001', 'Atty. Reyes', '2025-12-
01', 'no', NULL, NULL, 1, 5001, 0),
(1, '2026-01-15', 1002, '1985-08-12', 'CP-002', 'Maria Clara Reyes',
'Dr. Jose Rizal', '456 Mabini St, Quezon City', 'HCP-002', 'Reyes',
'Maria', 'Clara', 'SPA-002', 'P002', 'BK001', 'SR002', 'Atty. Cruz',
'2025-12-02', 'no', NULL, NULL, 1, 5002, 0),
(1, '2026-01-16', 1003, '1992-03-25', 'CP-003', 'Pedro Penduko', 'Dr.
Ana Lopez', '789 Bonifacio Ave, Cebu', 'HCP-003', 'Penduko', 'Pedro',
'Garcia', 'SPA-003', 'P003', 'BK002', 'SR003', 'Atty. Santos', '2025-12-
03', 'no', NULL, NULL, 2, 5003, 1),
(2, '2026-02-10', 1004, '1988-11-30', 'CP-004', 'Ana Patricia Gomez',
'Dr. Luis Torres', '321 Aguinaldo Rd, Davao', 'HCP-004', 'Gomez', 'Ana',
'Patricia', 'SPA-004', 'P004', 'BK002', 'SR004', 'Atty. Reyes', '2025-
12-10', 'yes', '2026-02-11', '10:30:00', 1, 5004, 1),
(2, '2026-02-10', 1005, '1995-07-14', 'CP-005', 'Carlos Miguel Tan',
'Dr. Elena Villanueva', '654 Luna St, Makati', 'HCP-005', 'Tan',
'Carlos', 'Miguel', 'SPA-005', 'P005', 'BK003', 'SR005', 'Atty. Dela
Rosa', '2025-12-15', 'no', NULL, NULL, 3, 5005, 2),
(2, '2026-02-11', 1006, '1982-01-08', 'CP-006', 'Sofia Isabelle Lim',
'Dr. Ramon Bautista', '987 Silang Ave, Pasig', 'HCP-006', 'Lim',
'Sofia', 'Isabelle', 'SPA-006', 'P006', 'BK003', 'SR006', 'Atty.
Mercado', '2025-12-20', 'no', NULL, NULL, 1, 5006, 0),
(3, '2026-03-05', 1007, '1998-09-18', 'CP-007', 'Rafael Antonio Uy',
'Dr. Patricia Ong', '159 Roxas Blvd, Paranaque', 'HCP-007', 'Uy',
'Rafael', 'Antonio', 'SPA-007', 'P007', 'BK004', 'SR007', 'Atty.
Garcia', '2026-01-05', 'yes', '2026-03-06', '14:15:00', 2, 5007, 1),
(3, '2026-03-05', 1008, '1991-12-03', 'CP-008', 'Gabriela Nicole Sy',
'Dr. Fernando Aquino', '753 Quezon Blvd, Taguig', 'HCP-008', 'Sy',
'Gabriela', 'Nicole', 'SPA-008', 'P008', 'BK004', 'SR008', 'Atty.
Ramos', '2026-01-10', 'no', NULL, NULL, 1, 5008, 0),
(3, '2026-03-06', 1009, '1987-04-22', 'CP-009', 'Miguel Alfonso Co',
'Dr. Teresa Javier', '246 Osmena St, Baguio', 'HCP-009', 'Co', 'Miguel',
```

```
'Alfonso', 'SPA-009', 'P009', 'BK005', 'SR009', 'Atty. Villar', '2026-
01-15', 'no', NULL, NULL, 3, 5009, 3),
(4, '2026-04-20', 1010, '1993-06-11', 'CP-010', 'Isabella Marie Chua',
'Dr. Roberto Domingo', '135 Kalayaan Ave, Pampanga', 'HCP-010', 'Chua',
'Isabella', 'Marie', 'SPA-010', 'P010', 'BK005', 'SR010', 'Atty.
Pineda', '2026-02-01', 'no', NULL, NULL, 1, 5010, 0),
(4, '2026-04-20', 1011, '1980-10-05', 'CP-011', 'Antonio Luis
Pangilinan', 'Dr. Carmen Salazar', '864 EDSA, Mandaluyong', 'HCP-011',
'Pangilinan', 'Antonio', 'Luis', 'SPA-011', 'P011', 'BK006', 'SR011',
'Atty. Hernandez', '2026-02-10', 'yes', '2026-04-21', '09:00:00', 2,
5011, 2);
```

OpenEdge (Progress) — `pub."id-print"` (11 records)

```
INSERT INTO pub."id-print" ("batch-no", "batch-date", "app-no", "dob",
"cp-no", "fullname", "hcp-fullname", "address", "hcp-cpno", "lname",
"fname", "mname", "spa-docno", "page-no", "book-no", "series-no", "spa-
atty", "spa-dt-issue", "printed", "printed-dt", "printed-tm", "id-type",
"id-code", "print-count")
VALUES
(1, '2026-01-15', 1001, '1990-05-20', 'CP-001', 'Juan Dela Cruz', 'Dr.
Maria Santos', '123 Rizal St, Manila', 'HCP-001', 'Dela Cruz', 'Juan',
'Santos', 'SPA-001', 'P001', 'BK001', 'SR001', 'Atty. Reyes', '2025-12-
01', 0, NULL, NULL, 1, 5001, 0),
(1, '2026-01-15', 1002, '1985-08-12', 'CP-002', 'Maria Clara Reyes',
'Dr. Jose Rizal', '456 Mabini St, Quezon City', 'HCP-002', 'Reyes',
'Maria', 'Clara', 'SPA-002', 'P002', 'BK001', 'SR002', 'Atty. Cruz',
'2025-12-02', 0, NULL, NULL, 1, 5002, 0),
(1, '2026-01-16', 1003, '1992-03-25', 'CP-003', 'Pedro Penduko', 'Dr.
Ana Lopez', '789 Bonifacio Ave, Cebu', 'HCP-003', 'Penduko', 'Pedro',
'Garcia', 'SPA-003', 'P003', 'BK002', 'SR003', 'Atty. Santos', '2025-12-
03', 0, NULL, NULL, 2, 5003, 1),
(2, '2026-02-10', 1004, '1988-11-30', 'CP-004', 'Ana Patricia Gomez',
'Dr. Luis Torres', '321 Aguinaldo Rd, Davao', 'HCP-004', 'Gomez', 'Ana',
'Patricia', 'SPA-004', 'P004', 'BK002', 'SR004', 'Atty. Reyes', '2025-
12-10', 1, '2026-02-11', '10:30:00', 1, 5004, 1),
(2, '2026-02-10', 1005, '1995-07-14', 'CP-005', 'Carlos Miguel Tan',
'Dr. Elena Villanueva', '654 Luna St, Makati', 'HCP-005', 'Tan',
'Carlos', 'Miguel', 'SPA-005', 'P005', 'BK003', 'SR005', 'Atty. Dela
Rosa', '2025-12-15', 0, NULL, NULL, 3, 5005, 2),
(2, '2026-02-11', 1006, '1982-01-08', 'CP-006', 'Sofia Isabelle Lim',
'Dr. Ramon Bautista', '987 Silang Ave, Pasig', 'HCP-006', 'Lim',
'Sofia', 'Isabelle', 'SPA-006', 'P006', 'BK003', 'SR006', 'Atty.
Mercado', '2025-12-20', 0, NULL, NULL, 1, 5006, 0),
(3, '2026-03-05', 1007, '1998-09-18', 'CP-007', 'Rafael Antonio Uy',
'Dr. Patricia Ong', '159 Roxas Blvd, Paranaque', 'HCP-007', 'Uy',
'Rafael', 'Antonio', 'SPA-007', 'P007', 'BK004', 'SR007', 'Atty.
Garcia', '2026-01-05', 1, '2026-03-06', '14:15:00', 2, 5007, 1),
(3, '2026-03-05', 1008, '1991-12-03', 'CP-008', 'Gabriela Nicole Sy',
'Dr. Fernando Aquino', '753 Quezon Blvd, Taguig', 'HCP-008', 'Sy',
'Gabriela', 'Nicole', 'SPA-008', 'P008', 'BK004', 'SR008', 'Atty.
Ramos', '2026-01-10', 0, NULL, NULL, 1, 5008, 0),
(3, '2026-03-06', 1009, '1987-04-22', 'CP-009', 'Miguel Alfonso Co',
'Dr. Teresa Javier', '246 Osmena St, Baguio', 'HCP-009', 'Co', 'Miguel',
```

```
'Alfonso', 'SPA-009', 'P009', 'BK005', 'SR009', 'Atty. Villar', '2026-
01-15', 0, NULL, NULL, 3, 5009, 3),
(4, '2026-04-20', 1010, '1993-06-11', 'CP-010', 'Isabella Marie Chua',
'Dr. Roberto Domingo', '135 Kalayaan Ave, Pampanga', 'HCP-010', 'Chua',
'Isabella', 'Marie', 'SPA-010', 'P010', 'BK005', 'SR010', 'Atty.
Pineda', '2026-02-01', 0, NULL, NULL, 1, 5010, 0),
(4, '2026-04-20', 1011, '1980-10-05', 'CP-011', 'Antonio Luis
Pangilinan', 'Dr. Carmen Salazar', '864 EDSA, Mandaluyong', 'HCP-011',
'Pangilinan', 'Antonio', 'Luis', 'SPA-011', 'P011', 'BK006', 'SR011',
'Atty. Hernandez', '2026-02-10', 1, '2026-04-21', '09:00:00', 2, 5011,
2);
```

Version: 1.0
Date: July 23, 2026
Purpose: Sync bridge between Progress OpenEdge and MySQL for Entrust Instant ID card printing. Reads `"id-print"` from OpenEdge and mirrors it to MySQL's `id_print` table.

Architecture Overview

```
┌──────────────────────────────────┐
│  Progress OpenEdge SQL Server    │  ← Source of truth (remote, provided
by client)
└──────────────┬───────────────────┘
               │ ODBC
┌──────────────▼───────────────────┐
│  OpenEdge ODBC Driver            │  ←
PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe
│  (DataDirect 8.0, 64-bit)        │
└──────────────┬───────────────────┘
               │
┌──────────────▼───────────────────┐
│  C# Windows Service              │  ← OpenEdgeBridge.exe (this
application)
│  (.NET 8, self-contained)        │
└──────────────┬───────────────────┘
               │ MySQL Connector
┌──────────────▼───────────────────┐
│  MySQL Server (XAMPP)            │  ← Local database, runs on the same PC
└──────────────┬───────────────────┘
               │ JDBC (mysql-connector-j-8.0.33.jar)
┌──────────────▼───────────────────┐
│  Entrust Instant ID              │  ← Card issuance system
└──────────────────────────────────┘
```

Data Flow

OpenEdge → MySQL (One-Way Sync)

```
OpenEdge "id-print" ──► C# Service ──► MySQL id_print table
       (read)              (sync loop)         (upsert)
```

All 24 fields from the `id-print` table are synced, including: - `batch-no`, `batch-date`, `app-no`, `dob` - `cp-no`, `fullname`, `hcp-fullname`, `address`, `hcp-cpno` - `lname`, `fname`, `mname` - `spa-docno`, `page-no`, `book-no`, `series-no`, `spa-atty`, `spa-dt-issue` - `printed`, `printed-dt`, `printed-tm` - `id-type`, `image-data`, `id-code`

Runs every N seconds (configurable in `appsettings.json`). Uses **INSERT ON DUPLICATE KEY UPDATE** to avoid duplicates. `print_count` is only inserted on first sync; subsequent updates do NOT overwrite it (Instant ID manages `print_count`).

> **Note:** `signature_data` and `qr_data` are **not** part of this sync — they are MySQL-only columns populated by Entrust Instant ID and never touched by the bridge (see Database Tables).

MySQL → OpenEdge (Printed Back-Sync)

```
MySQL id_print ──────► C# Service ──────► OpenEdge "id-print"
 (print_count > 0)    (check + mark)       (SET printed = yes)
```

When Instant ID increments `print_count > 0` on a record, the bridge detects it and sets `printed = yes` in the corresponding OpenEdge record.

Prerequisites

On the Target Windows PC

#	Component	Version / File	Notes
1	**Windows OS**	Windows 10/11 or Windows Server 2016+	64-bit required

#	Component	Version / File	Notes
2	**OpenEdge ODBC Driver**	`PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe`	Install first; configure a System DSN
3	**XAMPP**	Latest (8.x)	Provides local MySQL server
4	**MySQL Connector/J**	`mysql-connector-j-8.0.33.jar`	For Entrust Instant ID to connect to MySQL
5	**OpenEdgeBridge**	Published `publish/` folder	Self-contained, no .NET runtime needed

---

Local Development — Installing Progress OpenEdge Database

> **Use this section when no remote OpenEdge server is available.**
> You can install Progress OpenEdge locally, create a database, and run the SQL below.

Step 0-A: Download & Install Progress OpenEdge

1. Go to https://www.progress.com/openedge/downloads
2. Download **Progress OpenEdge Developer Studio** or **Progress OpenEdge Personal Edition** (free for development)
3. Run the installer (typically `PROGRESS_OE_12.x_WIN_64.exe`) as Administrator
4. Choose **Typical** or **Complete** installation
5. Default install path: `C:\Progress\OpenEdge\`

After installation, a **"Proenv"** shortcut will appear in your Start Menu. **Proenv** is a command prompt pre-configured with all OpenEdge environment variables.

Step 0-B: Create a Local OpenEdge Database

Open **Proenv** as Administrator and run:

```
:: Create a folder for your database
mkdir C:\OpenEdgeDatabases
cd C:\OpenEdgeDatabases

:: Create an empty database called 'iddb'
prodb iddb empty
```

This creates `iddb.db` and related files in `C:\OpenEdgeDatabases\`.

Step 0-C: Start the Database Server

In **Proenv**, run:

```
:: Start the database for multi-user access with both 4GL and SQL
enabled
proserve "C:\OpenEdgeDatabases\iddb.db" -S 11918 -ServerType Both -n 51
-L 8192
```

Parameter	Meaning
`-S 11918`	Service port (SQL and 4GL clients connect on this port)
`-n 51`	Maximum concurrent users
`-L 8192`	Lock table size (entries)
`-ServerType Both`	**Required** — enables both 4GL and SQL-92 engine

The database is now running. Keep this Proenv window open.

Step 0-D: Configure the ODBC System DSN

1. Press `Win + R`, type `odbcad32.exe`, press Enter
2. Go to the **System DSN** tab, click **Add...**
3. Select **Progress OpenEdge 12.x Driver** (or "DataDirect OpenEdge SQL Server")
4. Configure: - **Data Source Name:** `OpenEdgeInstantID` - **Description:** `OpenEdge Instant ID Database` - **Host Name:** `localhost` - **Port Number:** `11918` - **Database Name:** `iddb` - **User ID:** `admin`
5. Click **Test Connect** → should succeed
6. Click **OK**

Step 0-E: Create the Database Schema (`.df` File)

The table schema is defined in a `.df` (Data Definition) file. Create a file named `iddb.df` in `C:\OpenEdgeDatabases\` with the following content:

```
ADD TABLE "id-print"
  AREA "Schema Area"
  DUMP-NAME "id-print"

ADD FIELD "batch-no" OF "id-print" AS integer
  FORMAT "->,>>>,>>9"
  INITIAL "0"
  POSITION 2
  MAX-WIDTH 4
  ORDER 10

ADD FIELD "batch-date" OF "id-print" AS date
  FORMAT "99/99/9999"
  INITIAL ?
  POSITION 3
  MAX-WIDTH 4
  ORDER 20

ADD FIELD "app-no" OF "id-print" AS integer
  FORMAT "->,>>>,>>9"
  INITIAL "0"
  POSITION 4
  MAX-WIDTH 4
  ORDER 30

ADD FIELD "dob" OF "id-print" AS date
  FORMAT "99/99/9999"
  INITIAL ?
  POSITION 5
  MAX-WIDTH 4
  ORDER 40

ADD FIELD "cp-no" OF "id-print" AS character
  FORMAT "x(15)"
```

```
  INITIAL ""
  POSITION 6
  MAX-WIDTH 30
  ORDER 50

ADD FIELD "fullname" OF "id-print" AS character
  FORMAT "x(65)"
  INITIAL ""
  POSITION 7
  MAX-WIDTH 130
  ORDER 60

ADD FIELD "hcp-fullname" OF "id-print" AS character
  FORMAT "x(65)"
  INITIAL ""
  POSITION 8
  MAX-WIDTH 130
  ORDER 70

ADD FIELD "address" OF "id-print" AS character
  FORMAT "x(65)"
  INITIAL ""
  POSITION 9
  MAX-WIDTH 130
  ORDER 80

ADD FIELD "hcp-cpno" OF "id-print" AS character
  FORMAT "x(15)"
  INITIAL ""
  POSITION 10
  MAX-WIDTH 30
  ORDER 90

ADD FIELD "lname" OF "id-print" AS character
  FORMAT "x(55)"
  INITIAL ""
  POSITION 11
  MAX-WIDTH 110
  ORDER 100

ADD FIELD "fname" OF "id-print" AS character
  FORMAT "x(55)"
  INITIAL ""
  POSITION 12
  MAX-WIDTH 110
  ORDER 110

ADD FIELD "mname" OF "id-print" AS character
  FORMAT "x(55)"
  INITIAL ""
  POSITION 13
  MAX-WIDTH 110
  ORDER 120

ADD FIELD "spa-docno" OF "id-print" AS character
  FORMAT "x(25)"
  INITIAL ""
  POSITION 14
```

```
  MAX-WIDTH 50
  ORDER 130

ADD FIELD "page-no" OF "id-print" AS character
  FORMAT "x(20)"
  INITIAL ""
  POSITION 15
  MAX-WIDTH 40
  ORDER 140

ADD FIELD "book-no" OF "id-print" AS character
  FORMAT "x(20)"
  INITIAL ""
  POSITION 16
  MAX-WIDTH 40
  ORDER 150

ADD FIELD "series-no" OF "id-print" AS character
  FORMAT "x(20)"
  INITIAL ""
  POSITION 17
  MAX-WIDTH 40
  ORDER 160

ADD FIELD "spa-atty" OF "id-print" AS character
  FORMAT "x(65)"
  INITIAL ""
  POSITION 18
  MAX-WIDTH 130
  ORDER 170

ADD FIELD "spa-dt-issue" OF "id-print" AS date
  FORMAT "99/99/9999"
  INITIAL ?
  POSITION 19
  MAX-WIDTH 4
  ORDER 180

ADD FIELD "printed" OF "id-print" AS logical
  FORMAT "yes/no"
  INITIAL "no"
  POSITION 20
  MAX-WIDTH 1
  ORDER 190

ADD FIELD "printed-dt" OF "id-print" AS date
  FORMAT "99/99/9999"
  INITIAL ?
  POSITION 21
  MAX-WIDTH 4
  ORDER 200

ADD FIELD "printed-tm" OF "id-print" AS character
  FORMAT "x(35)"
  INITIAL ""
  POSITION 22
  MAX-WIDTH 70
  ORDER 210
```

```
ADD FIELD "id-type" OF "id-print" AS integer
  DESCRIPTION "1 = for RTC, 2=senior"
  FORMAT "->,>>>,>>9"
  INITIAL "0"
  POSITION 23
  MAX-WIDTH 4
  ORDER 220

ADD FIELD "image-data" OF "id-print" AS raw
  FORMAT "x(55)"
  INITIAL ""
  POSITION 24
  MAX-WIDTH 110
  ORDER 230

ADD FIELD "id-code" OF "id-print" AS integer
  FORMAT "->,>>>,>>9"
  INITIAL "0"
  POSITION 25
  MAX-WIDTH 4
  ORDER 240

.
PSC
cpstream=ISO8859-1
.
0000003166
```

Then, in **Proenv**, load the `.df` file into the database:

```
cd C:\OpenEdgeDatabases
prodict iddb
```

In the Data Dictionary menu, select **Admin → Load Data Definitions → .df file**, then choose `iddb.df`.

> **Note:** The database name is `iddb` and the schema resides under the `PUB` area (default).

Database Tables

`id-print` / `id_print`

> Converted from the provided `.df` schema.
> **Paste each CREATE TABLE directly into your SQL editor and run it.**

OpenEdge SQL — CREATE TABLE

```
CREATE TABLE "id-print" (
    "batch-no"      INTEGER       NOT NULL DEFAULT 0,
    "batch-date"    DATE          NULL,
    "app-no"        INTEGER       NOT NULL DEFAULT 0,
    "dob"           DATE          NULL,
    "cp-no"         VARCHAR(15)   NOT NULL DEFAULT '',
    "fullname"      VARCHAR(65)   NOT NULL DEFAULT '',
    "hcp-fullname"  VARCHAR(65)   NOT NULL DEFAULT '',
    "address"       VARCHAR(65)   NOT NULL DEFAULT '',
    "hcp-cpno"      VARCHAR(15)   NOT NULL DEFAULT '',
    "lname"         VARCHAR(55)   NOT NULL DEFAULT '',
    "fname"         VARCHAR(55)   NOT NULL DEFAULT '',
    "mname"         VARCHAR(55)   NOT NULL DEFAULT '',
    "spa-docno"     VARCHAR(25)   NOT NULL DEFAULT '',
    "page-no"       VARCHAR(20)   NOT NULL DEFAULT '',
    "book-no"       VARCHAR(20)   NOT NULL DEFAULT '',
    "series-no"     VARCHAR(20)   NOT NULL DEFAULT '',
    "spa-atty"      VARCHAR(65)   NOT NULL DEFAULT '',
    "spa-dt-issue"  DATE          NULL,
    "printed"       BIT           NOT NULL DEFAULT 0,
    "printed-dt"    DATE          NULL,
    "printed-tm"    VARCHAR(35)   NOT NULL DEFAULT '',
    "id-type"       INTEGER       NOT NULL DEFAULT 0,
    "image-data"    VARBINARY(110) NULL,
    "id-code"       INTEGER       NOT NULL DEFAULT 0
);
```

> **Note:** OpenEdge requires double-quotes around field names containing hyphens (`-`).
> The schema resides in the `PUB` area (default).
> `BIT` represents the OpenEdge `LOGICAL` type (yes/no → 1/0).
> `VARBINARY(110)` stores the raw image binary data (MAX-WIDTH 110 from the schema).

MySQL — CREATE TABLE

```
CREATE TABLE IF NOT EXISTS id_print (
    id              INT           NOT NULL AUTO_INCREMENT,
    batch_no        INT           NULL,
    batch_date      DATE          NULL,
    app_no          INT           NULL,
    dob             DATE          NULL,
    cp_no           VARCHAR(15)   NULL,
    fullname        VARCHAR(65)   NULL,
    hcp_fullname    VARCHAR(65)   NULL,
    address         VARCHAR(65)   NULL,
    hcp_cpno        VARCHAR(15)   NULL,
    lname           VARCHAR(55)   NULL,
    fname           VARCHAR(55)   NULL,
    mname           VARCHAR(55)   NULL,
```

```
    spa_docno      VARCHAR(25)   NULL,
    page_no        VARCHAR(20)   NULL,
    book_no        VARCHAR(20)   NULL,
    series_no      VARCHAR(20)   NULL,
    spa_atty       VARCHAR(65)   NULL,
    spa_dt_issue   DATE          NULL,
    printed        VARCHAR(3)    NULL        COMMENT 'yes / no (matches
OpenEdge LOGICAL FORMAT)',
    printed_dt     DATE          NULL,
    printed_tm     VARCHAR(35)   NULL,
    id_type        INT           NULL        COMMENT '1 = for RTC, 2 =
senior',
    image_data     LONGBLOB      NULL,
    signature_data LONGBLOB      NULL        COMMENT 'Signature image;
written by Entrust Instant ID (MySQL only, NOT synced from OpenEdge)',
    qr_data        LONGBLOB      NULL        COMMENT 'QR code image;
written by Entrust Instant ID (MySQL only, NOT synced from OpenEdge)',
    id_code        INT           NULL,
    print_count    INT           NULL        COMMENT 'Incremented by
Instant ID; triggers printed=yes back-sync',
    PRIMARY KEY (id),
    UNIQUE INDEX idx_app_no (app_no)
) ENGINE=InnoDB
  DEFAULT CHARSET=utf8mb4
  COLLATE=utf8mb4_unicode_ci;
```

> **Note:** `id` is an internal auto-increment primary key. `app_no` is the unique business key — the bridge uses it for deduplication (`ON DUPLICATE KEY UPDATE` fires when the same `app_no` is synced again).
>
> **MySQL-only columns (`signature_data`, `qr_data`):** These store the cardholder's signature and QR-code images and are **MySQL only**. They do **not** exist in the OpenEdge `"id-print"` table and are **never read from or written back to OpenEdge**. Entrust Instant ID populates them directly in MySQL. The bridge deliberately leaves them untouched — it does not include them in its INSERT/UPDATE, so syncing from OpenEdge will never overwrite them.
>
> **Already have the table?** If you created `id_print` before these columns existed, add them with: `sql ALTER TABLE id_print ADD COLUMN signature_data LONGBLOB NULL AFTER image_data, ADD COLUMN qr_data LONGBLOB NULL AFTER signature_data;`

Column Comparison — Plain English

Think of each row as **one person's ID-card application folder**. The columns are the details written on that folder.

OpenEdge name	MySQL name	Plain English meaning
`batch-no`	`batch_no`	**Group number** — which batch/run of applications this person came in with (like a shipment number)
`batch-date`	`batch_date`	**Date of that group** — when their batch was made
`app-no`	`app_no`	**Application number** — the unique file number of this person's ID application. This is the "fingerprint" that tells us which person is which
`dob`	`dob`	**Date of birth**
`cp-no`	`cp_no`	**The person's reference number** (their own control/card number)
`fullname`	`fullname`	**Full name** — complete name all in one
`hcp-fullname`	`hcp_fullname`	**Doctor's name** — the healthcare provider attached to this person
`address`	`address`	**Home address**
`hcp-cpno`	`hcp_cpno`	**Doctor's number** — the healthcare provider's own reference number
`lname`	`lname`	**Last name** (surname)
`fname`	`fname`	**First name**
`mname`	`mname`	**Middle name**

OpenEdge name	MySQL name	Plain English meaning
`spa-docno`	`spa_docno`	**Permission letter number** — the number of the Special Power of Attorney (SPA) document
`spa-atty`	`spa_atty`	**Lawyer's name** — the attorney who signed that permission letter
`spa-dt-issue`	`spa_dt_issue`	**Date the permission letter was issued**
`page-no`	`page_no`	**Page number** — which page in the record book this person is on
`book-no`	`book_no`	**Book number** — which record book
`series-no`	`series_no`	**Series number** — the serial number of the ID
`id-type`	`id_type`	**Kind of ID** — `1` = regular (RTC), `2` = senior citizen
`image-data`	`image_data`	**The person's photo** (picture file)
`id-code`	`id_code`	**ID code** — the code number printed on the card
`printed`	`printed`	**Has it been printed yet?** — `yes` or `no`
`printed-dt`	`printed_dt`	**Date it was printed**
`printed-tm`	`printed_tm`	**Time it was printed**
—	`signature_data`	**Person's signature picture** — *MySQL only*. Not in OpenEdge, not synced back. Entrust fills this in

OpenEdge name	MySQL name	Plain English meaning
—	`qr_data`	**QR code picture** — *MySQL only*. Not in OpenEdge, not synced back. Entrust fills this in
—	`print_count`	**How many times it got printed** — *MySQL only*. Instant ID counts this; it's never overwritten by the bridge
—	`id`	**Row number** — MySQL's own internal counter. Not business data, ignore it

30-second summary: The bridge copies the person's details (name, birthdate, address, photo, etc.) **from OpenEdge** → **MySQL**, like photocopying the folder. `app_no` is the **matchmaker** — it makes sure updates land on the right person. The three columns that exist **only in MySQL** (`signature_data`, `qr_data`, `print_count`) are filled in by the **Entrust machine**, and the bridge never touches them.

Step-by-Step Setup (Remote Server)

> **If you already have a remote OpenEdge server**, follow these steps.
> **If no server is available**, go back to Local Development — Installing Progress OpenEdge Database first.

1. Install OpenEdge ODBC Driver

Run the installer as Administrator:

```
PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe
```

After installation, open **ODBC Data Sources (64-bit)** and create a **System DSN**:

1. Press `Win + R`, type `odbcad32.exe`, press Enter
2. Go to the **System DSN** tab

3. Click **Add...**
4. Select **Progress DataDirect OpenEdge SQL Server** (or similar)
5. Configure: - **Data Source Name:** e.g., `OpenEdgeInstantID` - **Host:** (provided by your OpenEdge team, or `localhost` if local) - **Port:** (provided by your OpenEdge team, or `11918` if local) - **Database:** (provided by your OpenEdge team, or `iddb` if local) - **User ID / Password:** (provided by your OpenEdge team)
6. Click **Test Connect** to verify, then **OK**

2. Create the OpenEdge Tables

Connect to your OpenEdge database using any ODBC-compatible SQL client (DBeaver, HeidiSQL, etc.) and paste the SQL from the Database Tables section above:

- Run the `"id-print"` CREATE TABLE

3. Install XAMPP

1. Download from https://www.apachefriends.org/
2. Run the installer with default settings
3. Open the **XAMPP Control Panel**
4. Click **Start** next to **MySQL**
5. Click **Admin** to open phpMyAdmin

4. Create the MySQL Database & Tables

In phpMyAdmin (or any MySQL client), run:

```
CREATE DATABASE IF NOT EXISTS instantid_bridge
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_unicode_ci;

USE instantid_bridge;
```

Then paste and run the MySQL CREATE TABLE statements from the Database Tables section above:

- Run the `id_print` CREATE TABLE

5. Configure Entrust Instant ID (MySQL Connector/J)

1. Copy `mysql-connector-j-8.0.33.jar` to Entrust's `lib/` folder

2. Configure Entrust to point to the local MySQL database: - **Host:** `localhost` - **Port:** `3306` - **Database:** `instantid_bridge` - **User:** `root` (or a dedicated user) - **Table:** `id_print`

6. Deploy the Bridge Service

Copy the entire `publish/` folder to the target PC, e.g. `C:\OpenEdgeBridge\`.

Edit `appsettings.json`

```
{
  "ConnectionStrings": {
    "OpenEdge":
"DSN=OpenEdgeInstantID;Uid=your_user;Pwd=your_password;",
    "MySql":
"Server=localhost;Database=instantid_bridge;Uid=root;Pwd=your_mysql_pass
word;"
  },
  "Sync": {
    "IntervalSeconds": 10
  }
}
```

Setting	Description
`OpenEdge`	ODBC DSN name, user, and password
`MySql`	MySQL server, database, user, and password
`Sync:IntervalSeconds`	How often the sync loop runs (default: 10s)

7. Install as a Windows Service

Open **PowerShell as Administrator** and run:

```
New-Service -Name "OpenEdgeBridge" `
            -BinaryPathName "C:\OpenEdgeBridge\OpenEdgeBridge.exe" `
            -DisplayName "OpenEdge Bridge Service" `
            -Description "Syncs id-print data between OpenEdge and MySQL
for Entrust Instant ID" `
            -StartupType Automatic
```

```
Start-Service OpenEdgeBridge
```

Verify:

```
Get-Service OpenEdgeBridge
# Status should be: Running
```

Service Management Commands

```
Start-Service OpenEdgeBridge      # Start
Stop-Service OpenEdgeBridge       # Stop
Restart-Service OpenEdgeBridge    # Restart (after config changes)
Get-Service OpenEdgeBridge        # Check status

# Uninstall
Stop-Service OpenEdgeBridge
sc delete OpenEdgeBridge
```

Logging & Troubleshooting

Log Locations

Source	Location
File logs	`C:\OpenEdgeBridge\Logs\` (dated log files)
Windows Event Viewer	*Windows Logs* → *Application*, Source: `OpenEdgeBridge`

Common Issues

Problem	Likely Cause	Fix
Service starts then stops immediately	Wrong DSN name or credentials	Test the ODBC DSN in ODBC Administrator with **Test Connect**
"MySQL connection string not found"	`appsettings.json` not in same folder as `.exe`	Ensure `appsettings.json` is in `C:\OpenEdgeBridge\`
Duplicate rows in MySQL	Missing UNIQUE index	Run the `ALTER TABLE ... ADD UNIQUE INDEX` SQL above
OpenEdge ODBC driver not found	Wrong bitness (32 vs 64)	Use `odbcad32.exe` (64-bit), not the 32-bit one in SysWOW64
`Failure getting schema read lock (12138)`	SQL engine not initialized for database	1. Verify `-ServerType Both` or `-ServerType SQL` on proserve 2. In sqlexp run: `UPDATE TABLE STATISTICS AND INDEX STATISTICS AND ALL COLUMN STATISTICS;` 3. If still failing, restart DB: `proshut` then `proserve` with `-SQLTrc 3`
MySQL not accessible from Entrust	MySQL not running or wrong port	Check XAMPP Control Panel; MySQL must be **Started**
OpenEdge SQL CREATE TABLE fails with "Invalid table name"	Hyphens in table/field names not quoted	Wrap names containing hyphens in double quotes: `"id-print"`, `"batch-no"`
`image-data` field too small	VARBINARY length insufficient	The OpenEdge schema specifies MAX-WIDTH 110; adjust as needed

---

Sync Interval Tuning

The default sync interval is **10 seconds**. Adjust in `appsettings.json`:

```
"Sync": {
  "IntervalSeconds": 30
}
```

- **Lower** (e.g., 5s) = faster sync, more database load
- **Higher** (e.g., 60s) = slower sync, less load
- Recommended: 10–30 seconds for production

Port Summary

Component	Port	Direction
OpenEdge SQL Server	Provided by OpenEdge team	Outbound
MySQL (XAMPP)	3306	Local
Entrust → MySQL	3306	Local via JDBC

No inbound ports are opened by the bridge itself.

Upgrade Procedure

1. Stop the service: `Stop-Service OpenEdgeBridge`
2. Replace all files in `C:\OpenEdgeBridge\` with the new `publish/` output
3. Preserve your `appsettings.json` (do not overwrite with default)
4. Start the service: `Start-Service OpenEdgeBridge`

Checklist for Target PC

- [ ] Windows 10/11 or Server 2016+ (64-bit)

- [ ] OpenEdge ODBC driver installed and DSN configured
- [ ] XAMPP installed and MySQL running
- [ ] `instantid_bridge` database created in MySQL
- [ ] `id-print` / `id_print` table created in **both** OpenEdge and MySQL
- [ ] `mysql-connector-j-8.0.33.jar` placed in Entrust's lib folder
- [ ] Bridge `publish/` folder copied to `C:\OpenEdgeBridge\`
- [ ] `appsettings.json` updated with correct connection strings
- [ ] Windows Service installed and started
- [ ] Logs checked for errors